

Performance and cost of materials for lithium-based rechargeable automotive batteries

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It is widely accepted that for electric vehicles to be accepted by consumers and to achieve wide market penetration, ranges of at least 500 km at an affordable cost are required. Therefore, significant improvements to lithium-ion batteries (LIBs) in terms of energy density and cost along the battery value chain are required, while other key performance indicators, such as lifetime, safety, fast-charging ability and low-temperature performance, need to be enhanced or at least sustained. Here, we review advances and challenges in LIB materials for automotive applications, in particular with respect to cost and performance parameters. The production processes of anode and cathode materials are discussed, focusing on material abundance and cost. Advantages and challenges of different types of electrolyte for automotive batteries are examined. Finally, energy densities and costs of promising battery chemistries are critically evaluated along with an assessment of the potential to fulfil the ambitious targets of electric vehicle propulsion.

In retrospect, the years from 1900 to 1912 are remembered as the golden era of electric vehicles (EVs)¹. In 1912, the number of EVs in use amounted to 30,000 in the United States¹. Most of these EVs were powered by a lead–acid battery (LAB), which had been invented by Sinsteden (1854) and Planté (1859); each cell delivered a cell voltage of around 2 V (ref. ²). Due to the poor active mass utilization and unfavourable charge–discharge mechanism in LABs³, their practical energy content only amounts to 40 Wh kg^{−1} and 90 Wh l^{−1}, respectively⁴. Their Coulombic efficiency and energy efficiency are also only moderate and sum up to 80% and 70%, respectively². Today, nearly all of the 1.3 billion vehicles on the road worldwide use LABs for starting, lighting and ignition (SLI) applications. After a century of combustion engine supremacy, the imperative to decarbonize the transport sector is expected to lead to another golden era of electric mobility: in 2016, 160,000 plug-in hybrid electric vehicles (PHEVs) were sold in the US alone⁵.

Nickel–metal hydride (NiMH) batteries have been the primary choice for hybrid electric vehicles (HEVs)⁴. NiMH cells have a nominal cell voltage of 1.2 V and can deliver an energy content of up to 80 Wh kg^{−1} and 250 Wh l^{−1}, respectively². However, Coulombic efficiency (70%) and energy efficiency (65%) are even lower than in LABs². Nowadays, PHEVs and battery-electric vehicles (BEVs) solely use lithium-ion batteries (LIBs) for propulsion⁴.

Next to their higher energy contents of up to 260 Wh kg^{−1} and 700 Wh l^{−1} at cell level, as well as higher Coulombic (99%) and energy efficiencies (up to 95%)^{6,7}, the main advantage of LIBs lies in the possibility of designing a vast range of different power to energy ratios (P:E), such as P:E = 15, 8 or 3 for HEVs, PHEVs or BEVs, respectively^{6,8,9}. To achieve mass market penetration, the US Department of Energy and the Advanced Battery Consortium estimated that a driving range of at least 500 km is required for BEVs, which corresponds to approximately 235 Wh kg^{−1} and 500 Wh l^{−1} at battery pack level and 350 Wh kg^{−1} and 750 Wh l^{−1} at cell level^{5,8,10}. Further, the pack cost needs to fall below 125 US\$ kWh^{−1} (ref. ⁵). It should be noted that there are other more optimistic views on energy and cost. State-of-the-art automotive LIB packs show up to 130–140 Wh kg^{−1} and over 210 Wh l^{−1}, respectively⁸.

To meet energy and cost targets, improvements along the whole battery value chain are needed (Fig. 1a). Besides LIBs, lithium metal-based batteries with solid electrolytes are also considered for EV application. The cell design concepts of LIB cells with two insertion electrodes and of solid-state lithium metal-based cells are depicted in Fig. 1b. This Review surveys the state-of-the-art and recent advances in active materials, electrolytes, separators and cell chemistries for automotive batteries. We focus on three most important aspects: performance, production and cost. We also give a perspective on future developments at the end.

Anode materials

With respect to anode chemistries, carbonaceous materials, in particular synthetic and artificial graphites (SGs) and natural graphites (NGs) as well as amorphous (hard and soft) carbons, can be considered as state-of-the-art negative electrode materials^{4,8,11}. With respect to EV applications, and compared to NGs, SGs show outstandingly high levels of purity and less fluctuating quality¹². Quite often, mixtures of amorphous and graphitic carbons are used to optimize the different characteristics with respect to application, for example, to optimize the P:E ratio^{4,13}. Currently, in some commercial cells (for example, Panasonic or Hitachi Maxell–Kopin), small amounts of silicon (mostly as SiO_x) are added to the carbon anode to further enhance cell energy¹¹. Also, lithium titanate (LTO) is utilized in commercial cells (for example, Toshiba's SciB)¹¹. However, due to their low resulting cell voltage and high-power capability, LTO-based cells are more suitable for power applications, particularly for recuperation such as in electric buses; thus, they are not further discussed in this work. Lithium metal is considered as the most promising future anode material, in particular for application in all-solid-state batteries (ASSBs) using ceramic or polymeric electrolytes; currently it is already practically used in Li-metal polymer batteries by Bolloré¹⁴. The current market share of 43% of SG and 46% of NG (2016 data¹⁵) for LIBs, with only 7% for amorphous carbon, clearly evidences the dominance of carbon-based anode materials. In contrast, LTO and Si-based anode materials only make up around 2% each¹².

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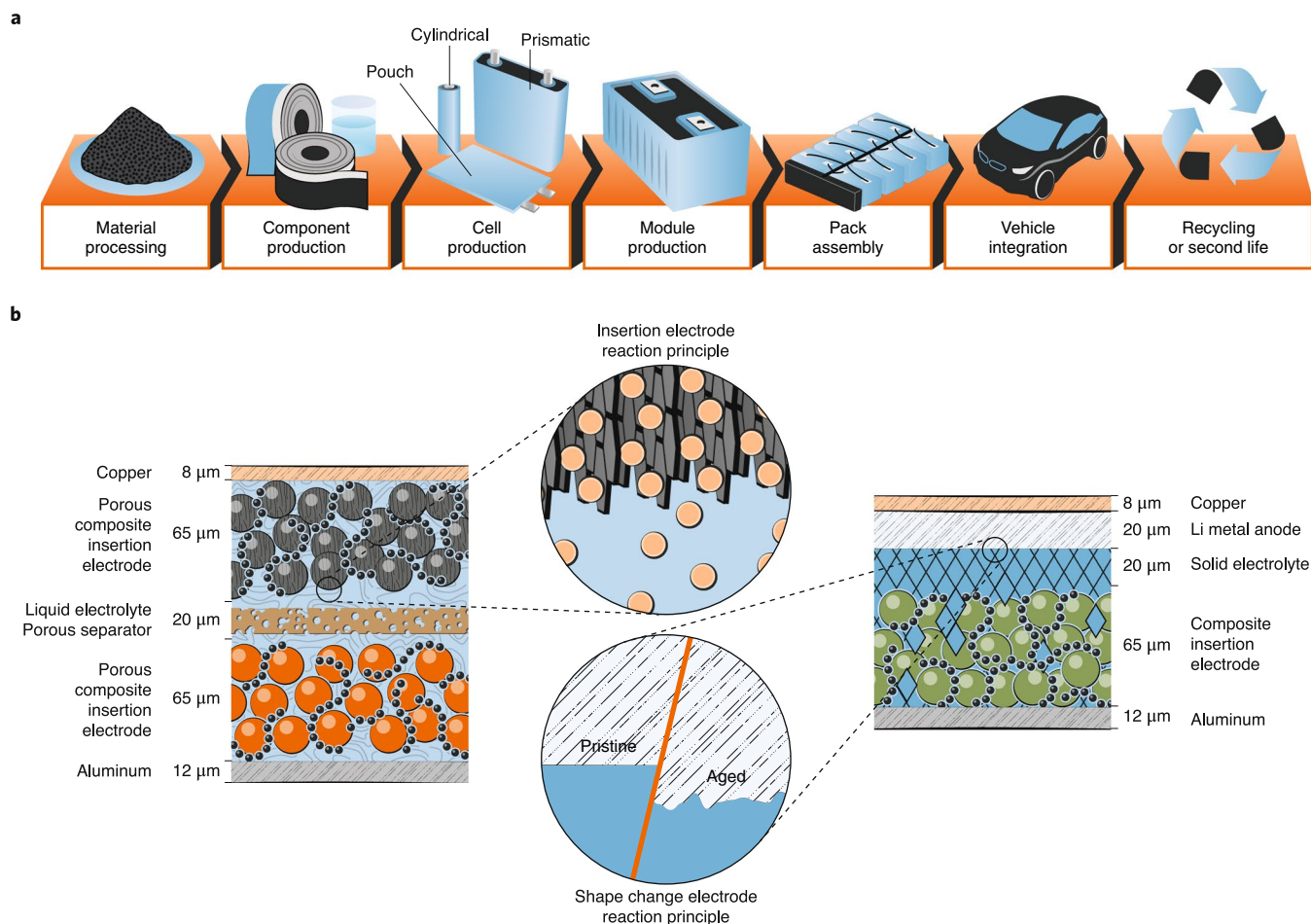


Fig. 1 | Battery value chain and lithium ion versus Li-metal cell chemistries. a, Value chain of rechargeable batteries for electric mobility. **b**, Schematic cross-sections of a lithium-ion battery (LIB, left) and an all-solid-state battery (ASSB) cell configuration (right). Thickness data refer to state-of-the-art lithium-ion batteries and to suitable ASSBs (while even lower values for the solid electrolyte and Li-metal anode are targeted).

Figure 2 displays the production processes for state-of-the-art graphite-based anode materials, for Li-metal foil as well as possible commercial routes for production of Si-C and SiO_x-Si-C composites. It is difficult to get detailed information from material manufacturers for the latter processes. The processing of SG (Fig. 2a) comprises of three major steps. First, the precursor material, for example, a type of coke, is transferred into an intermediate material, the so-called soft (graphitizable) carbon, by calcination at 800–1,200 °C. The soft carbon is further treated by crushing and grinding, and the particles are classified according to their particle size (range in μm). The material is graphitized in a second step at temperatures higher than 2,500 °C for several days to weeks including cooling phases. In the final step, typically mechanical particle refinement is conducted by grinding, classifying and eventually coating to optimize particle design^{13–17}. The production of NG (Fig. 2b) starts with the mining from graphite ores and several steps for separation and concentration of the flaky graphite particles, particularly by flotation^{15,16,18,19}. Thereby, purity levels of around 95% can be reached, with the main impurities being silicate mineral grains and chemical compounds of S, K, Na, and so on. Then, graphite flakes are treated by sophisticated milling techniques to obtain spherical particles in the μm range (via micronization and spheroidization)^{14,15,19}. The mechanical treatment is followed by wet-chemical and/or thermal purification processes, most likely leaching with hydrofluoric acid or thermal treatment at over 2,000 °C to decrease the impurity content to less than 500 ppm¹⁶. Similar to SG processing, the final

step focuses on particle refinement. The carbon coating process is a highly important step to ensure good performance¹⁶. The price for NG amounts to around 8 US\$ kg⁻¹, while as of 2016 SG had a higher price of around 13 US\$ kg⁻¹. This is mostly related to the energy cost during graphitization¹². NG has been categorized as a critical, strategic material by several governments including in the US and Europe; carbons and graphites can, however, always be synthesized. In particular, the price for SG is expected to decrease significantly (to less than 10 US\$ kg⁻¹ in 2025), which is related to the increasing demand and possible supply bottlenecks for NG.

Even though various reports for lab-scale preparation of Si-C and SiO_x-Si-C composite anode materials have been published, there is not much information about commercial production processes publicly available. It is known that so far only a small mass fraction of Si or SiO_x (<5 wt%) is added to the anode¹¹. We present two practically suitable production processes for Si-based anode materials (Fig. 2c). Si nanoparticles are typically prepared by plasma gas phase synthesis, for example, by pyrolysis of silanes²⁰, while SiO_x particles are obtained by reduction of SiO or SiO₂ precursors, which can be performed by milling or gas phase reactions^{21,22}. In the following step, the Si-based active materials have to be embedded in a carbon matrix, for example by mixing with graphite or carbon precursors followed by additional carbonization and milling procedures. In the final step, the composite particles can be further optimized, for example by carbon coating to decrease the specific surface area and, thus, reaction with the electrolyte¹⁶.

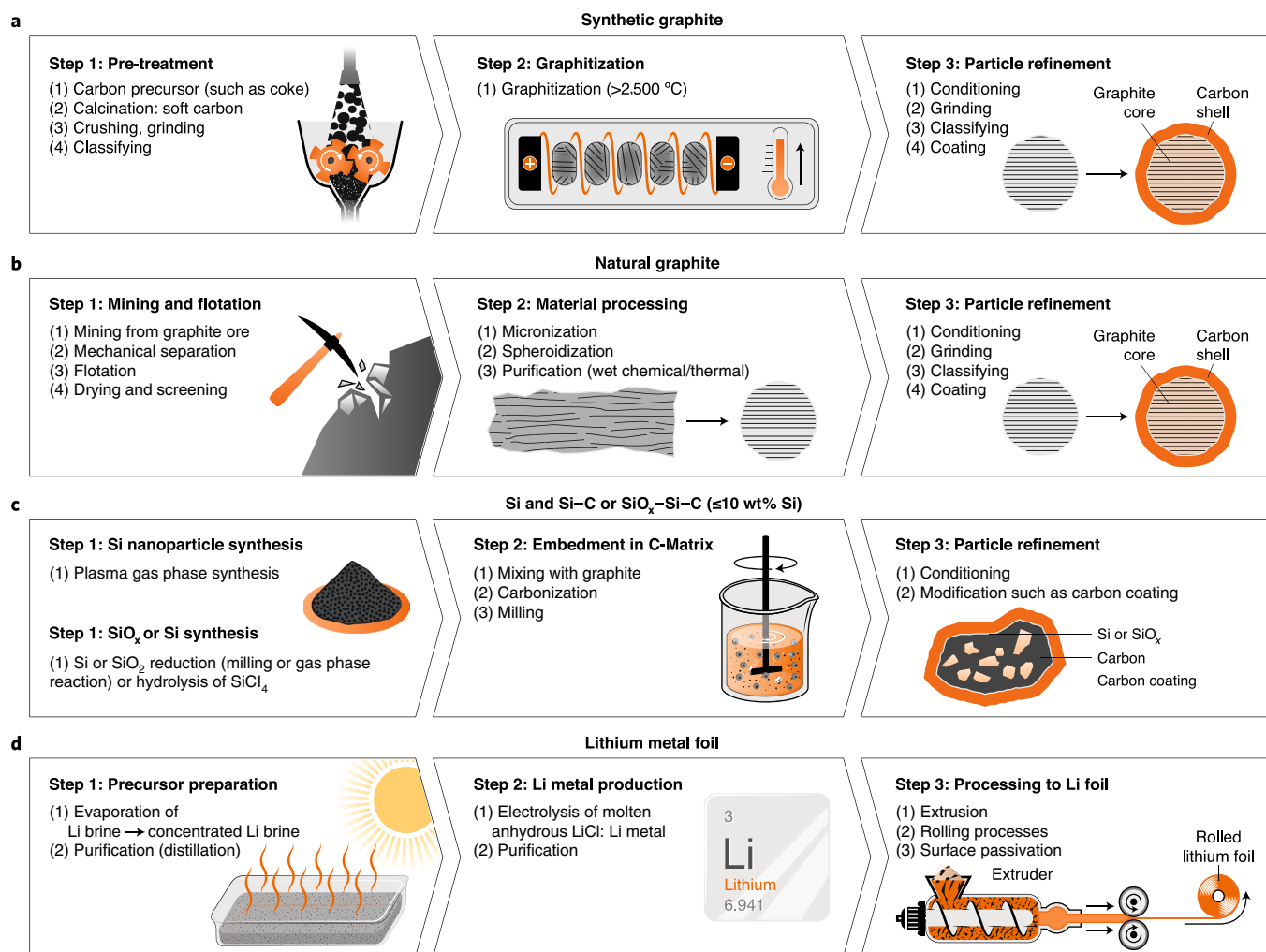


Fig. 2 | Production processes for negative electrode materials for LIB and Li-metal cells. a–d, Schematic illustration of industrial syntheses of synthetic graphite/soft carbon (**a**), natural graphite (**b**), Si-C or SiO_x-Si-C composite materials (**c**) and Li-metal (**d**).

The production of Li-metal can start from several natural raw materials, but most often from brines, which are first concentrated via evaporation and converted to LiCl before purification (Fig. 2d). Highly purified molten anhydrous LiCl is processed via electrolysis to Li metal²³. Distillation results in the removal of low-melting-point impurities such as Na. In the next step, Li-metal is shaped, for example, into foil by extrusion and/or rolling, with purity and thickness being the most important characteristics^{17,24,25}. Finally, Li-metal foil is typically stabilized by surface passivation, for example, by gas treatment, by coating with a layer of polyethylene wax or by surface fluorination²⁶. Recent price fluctuations for metallic lithium has led to cost concerns for Li-metal batteries such as ASSBs^{25,27}. Considering a Li-metal ingot price of 50–130 US\$ kg⁻¹ (based on Li in the raw material Li₂CO₃ with a price of 10–25 US\$ kg⁻¹, using a Li₂CO₃ to Li-metal conversion factor of 5.3, which accounts for the molar lithium content within Li₂CO₃)^{28,29}, and costs for processing to thin foils, one may easily end up with a Li-foil price of 300–400 US\$ kg⁻¹. However, it has also been communicated that processing to thin foils (much thinner than 100 μm) can even lead to costs of over 1,000 US\$ kg⁻¹ depending on the processing technique (evaporation, vapour deposition, extrusion and/or rolling) and the targeted Li-foil thickness. These costs are higher than LIB materials (much less than 100 US\$ kg⁻¹) and Li-metal based chemistries may have an uncertain future, unless the costs of Li-metal and its

processing decrease significantly in the future and inexpensive shaping technologies for the Li-metal electrode are developed^{30–32}.

Today, a trend to gradually increase the silicon content in graphite or carbon-based anodes is observed, while there might be a sharper future technology change going from LIBs to Li-metal-based ASSBs. The performance characteristics of anode materials, including specific and volumetric capacities as well as energy efficiencies, are summarized in Fig. 3a. The challenges of Si-based anode materials, such as volume change upon de-/lithiation and related mechanical degradation and active lithium loss^{33–36}, as well as safety issues of lithium metal anodes^{37,38} have been reported widely. Furthermore, various concepts to enhance the mechanical stability of Si-based composite anodes^{22,33,34} and to mitigate active lithium loss (for example, by pre-lithiation)^{36,39} in LIBs as well as to stabilize the Li-metal surface for ASSBs have been presented^{37,38,40}.

Cathode materials

Since the commercialization of LIBs, the cathode has proven a bottleneck with regard to specific capacities. Key requirements for positive active materials for automotive batteries include high specific and volumetric capacities and high discharge potentials versus Li/Li⁺, high intrinsic safety, high tap density, fast kinetics and good capacity retention. Due to their high technological maturity, LiMO₂-type layered oxide cathodes, which incorporate transition metals (M) such as nickel, cobalt and manganese (NMC) or nickel,

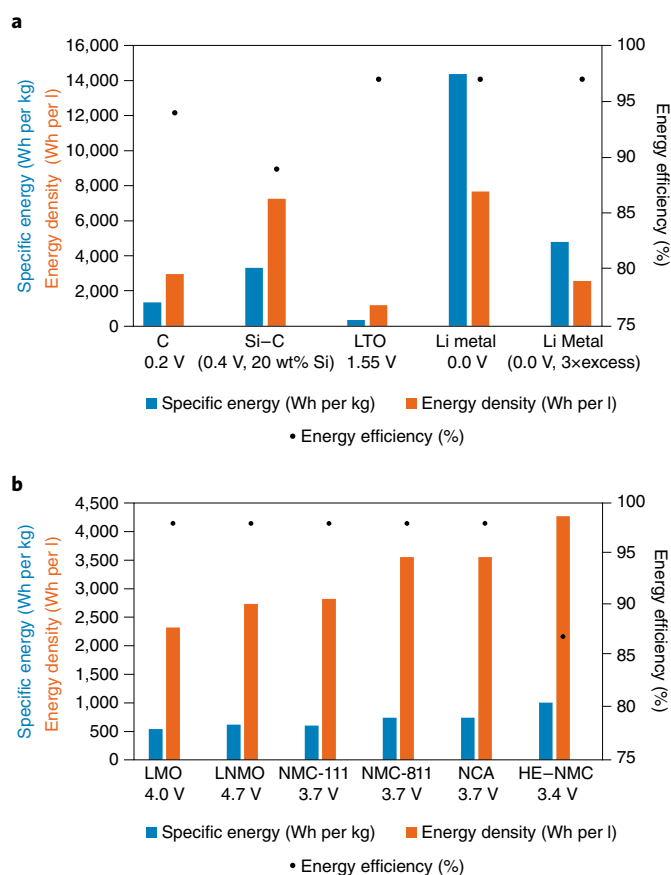


Fig. 3 | Energy densities and energy efficiencies of electrode materials at the material level. a, b, Specific energies, energy densities and average energy efficiencies of negative (**a**) and positive (**b**) active materials at material level¹⁷. The energy density calculations are based on average discharge potentials (versus Li/Li⁺) of the respective active material given below each bar. The energy content was calculated using a spreadsheet-based calculation tool, using material densities and electrochemical properties (such as practical discharge potential and redox potential) as input parameters. The calculations assume a NMC positive electrode with a redox potential of 3.7 V versus Li/Li⁺ (**a**) and a graphitic negative electrode with a redox potential of 0.2 V versus Li/Li⁺ (**b**). The energy efficiency (defined as discharge energy divided by charge energy⁷) values refer to electrodes with typical electrode compositions and active material particles sizes of commercial batteries at a current rate of 1C.

cobalt and aluminium (NCA), are currently most widely used as positive active materials for automotive batteries^{8,41}.

In contrast to the electrochemically inactive Mn⁴⁺ ions and the one-electron process of the Co^{3+/4+} redox couple, a part of the nickel (depending on the Mn content) offers a two-electron redox process (Ni^{2+/4+}) in NMC-type oxides. In general, a higher Ni content implies a higher capacity. A widely adapted strategy to get the most out of the energy content of NMC positives therefore lies in maximizing the Ni content (Fig. 3b)⁴². Under this strategy, the specific capacity can be increased from around 160 mAh g⁻¹ for NMC-111 (LiNi_{1/3}Co_{1/3}Mn_{1/3}O₂) to more than 200 mAh g⁻¹ for NMC-811 (LiNi_{0.8}Co_{0.1}Mn_{0.1}O₂) at an almost unchanged redox potential. Ni-rich layered cathodes such as NMC-811 or NCA (LiNi_{0.8}Co_{0.15}Al_{0.05}O₂), however, currently have a significantly shorter cycle life and a lower thermal stability and so are less safe⁴¹. Nevertheless, layered oxide cathodes with Ni contents exceeding 80% are currently under development⁴¹. Comparing NCA and NMC-811, which provide very similar practical capacities, NCA shows advantages with regard to

capacity retention, a smaller tendency for transition metal dissolution (no Mn), and power, while NMC-811 shows more favourable properties with regard to thermal stability (more inactive Mn). Ni-rich cathodes exhibit a higher reactivity towards moisture and the battery electrolyte, which may require additional measures during synthesis, electrode processing and cell operation. Besides the optimization of the elemental composition, material coating and the choice of adequate electrolyte formulations are key considerations for these cathode materials⁴¹.

A promising approach to increase the energy density of layered oxides, while preserving cycle life and safety, has been established⁴¹, involving the synthesis of spherical NMC particles with a Ni-rich core and Mn-rich shell. While the particle core provides high capacity, the Mn-enriched shell exhibits an enhanced stability versus the electrolyte. Since its initial report in 2005⁴³, this concept has been continuously optimized with full concentration-gradient NMC (FCG-NMC) particles with Ni contents greater than 80% and, thus, specific capacities beyond 220 mAh g⁻¹ (ref. ⁴⁴). Despite its high average nickel content, the capacity retention of FCG-NMC is claimed to exceed that of NCA⁴⁴.

Regardless of these promising developments, the maximum theoretical energy contents of around 1 kWh kg⁻¹ and 5 kWh l⁻¹ at material level could mark the ultimate physicochemical limit of insertion cathode materials involving a single electron transfer per transition metal^{45,46}.

Li-rich and Mn-rich oxides (LMR-NMC) constitute a structurally related materials class that provides exceptionally high specific capacities of up to 300 mAh g⁻¹ at a slightly lower discharge potential than conventional NMC⁴⁷. A higher Mn content of LMR-NMC cathodes results in a significant cost advantage over conventional NMC, but also leads to the conversion from the layered to spinel structure upon cycling⁴⁸. This conversion causes voltage and capacity fade, which are most critical performance issues⁴⁹. Further down-sides of LMR-NMC lie in its lower crystallographic density, slow kinetics, and the voltage hysteresis caused by the anionic redox activity^{48,50}.

Figure 4 illustrates the synthesis processes of transition metal oxides⁵¹ as well as of lithium transition metal phosphates⁵². Layered oxides with mixed transition metal composition are typically prepared over a co-precipitation route. An aqueous solution of the transition metal salts (for example, MSO₄) and a base (NaOH or Na₂CO₃) as well as a surfactant (NH₄OH) are pumped into a continuously stirred tank reactor (CSTR), in which the metal hydroxide or carbonate precipitates. After filtering and repeatedly washing to remove impurities, the material is dried and sieved before it is mixed with a stoichiometric amount of a lithium salt. While Li₂CO₃ is significantly cheaper, Ni-rich cathodes, such as NMC-811, have more favourable physical and electrochemical properties when calcined with LiOH (ref. ⁵³). The desired product is formed through calcination at 650–950 °C. The as-prepared cathode powder is then deagglomerated and may optionally be surface treated to stabilize the electrode/electrolyte interface and, thus, electrochemical performance^{41,54}.

Lithium metal phosphates such as LFP are accessible through numerous synthesis routes. Industrial production, however, often takes a two-step route: the first step involves the preparation of the precursor, which may be prepared through spray-drying of an aqueous solution, already containing all required reactants⁵⁵. The implementation of carbon is crucial to overcome the intrinsically low electronic conductivity of this materials class⁵⁶. The desired product is finally obtained through calcination within an inert or mildly reducing atmosphere.

The cathode active material accounts for more than 20% of the cost of current NMC-based LIBs, while raw material makes up more than 50% of the cathode cost⁵¹. The world's cobalt reserves are geographically highly concentrated in politically instable regions in

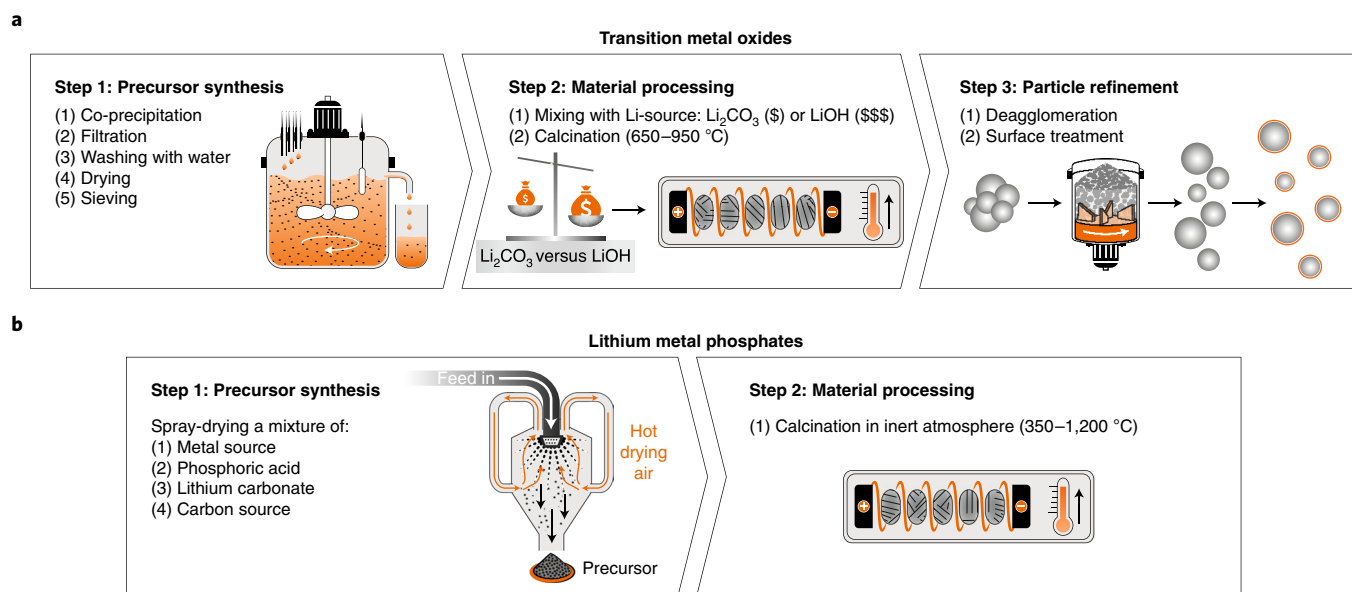


Fig. 4 | Production processes for positive electrode materials for LIB cells. **a, b**, Schematic illustration of industrial syntheses of lithium transition metal oxides: layered oxides LiMO_2 , spinel oxides LiM_2O_4 or LMR-NMC $\text{Li}_x\text{M}_{1-x}\text{O}_2$ with $\text{M} = \text{Ni}, \text{Co}, \text{Mn}, \text{Al}$ and $0 < x < 0.2$ (**a**), and lithium transition metal phosphates: LiMPO_4 with $\text{M} = \text{Fe}, \text{Ni}, \text{Co}, \text{Mn}$ (**b**).

Central Africa, allegedly involving child labour and severe environmental damage. The cobalt price has undergone strong fluctuations within the last decade and has recently risen markedly⁵⁷. The chosen transition metal composition of the cathode therefore not only has a large impact on energy density, cycle life and safety, but also on the battery cost⁵⁸. Due to the high cobalt price, cobalt-free positives such as the high-voltage spinel oxide $\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4$ (LNMO), $\text{LiFe}_{1-x}\text{Mn}_x\text{PO}_4$ and Co-free LMR-oxides are being discussed as potential low-cost, environmentally benign alternative positives⁴⁵. Despite its low energy content (see Fig. 3b) and its limited cycle life (particularly at elevated temperatures), the cubic spinel oxide LiMn_2O_4 (LMO) is also still used in many of today's EV batteries. LMO has an excellent rate capability at low cost and is, thus, often added in significant amounts to cathode material blends (see Table 1)⁸.

Global lithium reserves may be considered less critical, although lithium brines are geographically concentrated in South America and China⁵⁹. Lithium prices have been continuously on the rise in past years, mainly due to the fast-growing demand of the battery market, which currently consumes around 40% of worldwide production⁶⁰. It should be noted that the commonly quoted 'lithium price' often refers to lithium carbonate. Accordingly, the price of refined pure Li metal is significantly higher. In view of the expected further growth of the battery market, an increase of the currently negligible recycling quota for lithium should be strived for to make electric mobility more sustainable. Global nickel reserves are geographically less concentrated and sufficiently large to sustain future cathode material production. Today, the majority of nickel consumption goes to steel production and only a minor share falls into battery materials. Despite its high economic importance, no supply risk is presently to be expected for nickel⁶¹. Reserves of further battery-relevant transition metals such as iron, aluminium and manganese are non-critical, so these non-toxic elements are available at very low cost.

Electrolytes and separators

Commercial LIBs typically contain electrolytes (still almost exclusively) based on lithium hexafluorophosphate (LiPF_6) as conducting salt that is dissolved in mixtures of cyclic and linear organic carbonate solvents^{62,63}. The salt concentration is typically in the range of

0.8–1.5 mol L^{-1} in order to achieve maximum ionic conductivity⁶². To tailor the performance of LIBs with regard to a desired functionality, electrolyte additives are added in small concentrations of less than 5 wt% to leave the bulk electrolyte properties unchanged⁶⁴. The most commonly used cyclic carbonate is ethylene carbonate (EC)⁶⁵. EC is produced commercially via the catalytic oxidation of ethylene with oxygen to yield ethylene oxide, which is then reacted with CO_2 in the presence of an anion exchange resin catalyst (Fig. 5a). The production of the linear carbonate dimethyl carbonate (DMC) is based on oxidative carbonylation of methanol^{66–68}. DMC is reacted with ethanol in a transesterification to form diethyl carbonate (DEC)⁶⁹. The two symmetrical linear organic carbonates DMC and DEC are reacted in the presence of a catalyst to form ethyl methyl carbonate (EMC) in a disproportionation reaction⁷⁰. Organic carbonates are high-volume commodity chemicals that are derived by a C1 or C2 route (Fig. 5a) from cracker reactors. Beyond their relatively small-volume use in (electro-)chemical applications, they are applied as solvents in industrial cleaners, cosmetics, paint strippers, and lubricants, as well as plasticizers and intermediates for production of polycarbonates. In contrast, lithium salts are fine chemicals, only used in electrochemical (and chemical) applications. The lithium salt, mainly LiPF_6 , is the main cost driver in carbonate-based electrolytes. LiPF_6 is produced by reacting lithium fluoride with phosphorous pentachloride in anhydrous hydrogen fluoride (AHF). Thereby, PF_5 is generated in situ. LiPF_6 is further purified with additional AHF (Fig. 5b)⁷¹. Thus, LiPF_6 in various purity classes is achieved, starting from 99.0% for industrial grade to 99.99% for EV plus grade. Depending on the purity and composition, prices for carbonate-based electrolytes range from 7–20 US\$ kg^{-1} (ref. 72).

Recent studies verified, contrary to common belief in literature, that these state-of-the-art electrolytes are oxidatively quite stable up to 5.0 V or higher, versus Li/Li^+ at 20 °C, and thus can be used in combination with high voltage cathode materials⁷³. The main challenge of these electrolytes is their poor intrinsic safety, especially at elevated temperatures (60 °C). Linear carbonates are highly volatile and flammable, for example, DMC has a flash point of only 18 °C (refs 65,74). Further, LiPF_6 is highly susceptible towards hydrolysis and decomposes under formation of highly toxic organophosphates and highly corrosive hydrogen fluoride⁷⁵.

Table 1 | Overview of past and future LIB cell chemistries and characteristics of cells and batteries for EVs

Cell format	Cell chemistry		Cell configuration and characteristics				Battery size and range		Cell use
	Anode	Cathode	Capacity (Ah)	Voltage (V)	Specific energy (Wh kg ⁻¹)	Energy density (Wh l ⁻¹)	Energy (kWh)	Driving range (km)	OEM and model ^a
Prismatic cells									
Li Energy Japan	C	LMO-NMC	50	3.70	109	218	16	160	Mitsubishi i-MIEV (2008)
Toshiba	LTO	NMC	20	2.30	89	200	20	130	Honda Fit EV (2013)
Samsung SDI	C	LMO-NMC	63	3.65	172	312	24	140	Fiat 500e (2013)
Samsung SDI	C	LMO-NCA-NMC	60	3.70	122	228	22	130	BMW i3 (2014)
Panasonic/ Sanyo	C	NMC	25	3.70	130	215	24	190	VW e-Golf (2015)
Samsung SDI	C	LMO-NCA-NMC	37	3.70	185	357	36	300	VW e-Golf (2016)
Samsung SDI	C	LMO-NCA-NMC	94	3.70	189	357	33	183	BMW i3 (2017)
Pouch cells									
AESC	C	LMO-NCA	33	3.75	155	309	24	135	Nissan Leaf (2010)
A123	C	LFP	20	3.30	131	247	21	130	Chevrolet Spark EV (2012)
LG Chem	C	LMO-NMC	16	3.70	-	-	35.5	160	Ford Focus EV (2012)
LG Chem	C	LMO-NMC	36	3.75	157	275	26	150	Renault Zoe (2012)
Li-Tec	C	NMC	52	3.65	152	316	17	145	Smart Fortwo EV (2013)
SK Innovation	C	NMC	38	3.70	-	-	27	145	Kia Soul EV (2014)
AESC	C	LMO-NCA	40	3.75	167	375	30	172	Nissan Leaf (2015)
LG Chem	C	NMC	56	3.65	186	393	60	383	Chevrolet Bolt (2016)
LG Chem	C	NMC	59	3.70	241	466	41	400	Renault Zoe (2017)
Cylindrical cells									
Panasonic	C	NCA	3.2	3.60	236	673	60–100	330–500	Tesla S (2012)
Panasonic	Si or SiO–C	NCA	3.4	3.60	236	673	60–100	330–500	Tesla X (2015)
Panasonic	Si–C or SiO _x –C	NCA	4.75	3.60	260	683	75–100	490–630	Tesla 3 (2017)

^aYear in brackets indicates start of production. C, graphite; Si, silicon; LTO, Li₄Ti₅O₁₂; LMO, LiMn₂O₄; LFP, LiFePO₄; NMC, LiNi_{1–x-y}Co_xMn_yO₂; NCA, LiNi_{0.8}Co_{0.15}Al_{0.05}O₂; OEM, original equipment manufacturer. Data from refs ^{4,11} and ⁸².

The use of a liquid organic electrolyte within the LIB inevitably requires a porous membrane as separator, placed between the negative and positive electrode to prevent electrical short circuits⁷⁶. The annual separator demand in LIB cells added up to roughly 1.3×10^9 m² in 2016 (ref. ¹²). Technically, four basic separator types are currently available, differing in product chemistry and production process: (1) microporous membranes, (2) non-woven mats, (3) ceramic-coated separators and (4) solid inorganic, polymeric or hybrid electrolytes, where the latter combine the electrolyte and separator function in one⁷⁶. The majority of the microporous separators are based on polyolefins (PO: polypropylene, polyethylene)

in single- or multi-layer design⁷⁶. Single-layer separators are either produced in a dry process (Fig. 5c), starting from melt extrusion, annealing and stretching (mono- or bi-axial), or in a wet process, starting from a polymer solution, extrusion, solvent extraction and stretching, at production cost often below 0.6 US\$ m⁻² (ref. ⁷⁶). Non-woven mats are the separator of choice for lead–acid batteries⁷⁶, but are frequently used in LIB-related research. However, traditional manufacturing processes such as melt blowing result in large pore size, inappropriate for practical LIBs⁷⁶. Therefore, non-woven separators for use in LIBs are produced by electrospinning or additional coating⁷⁶. However, production costs exceed the PO separator costs

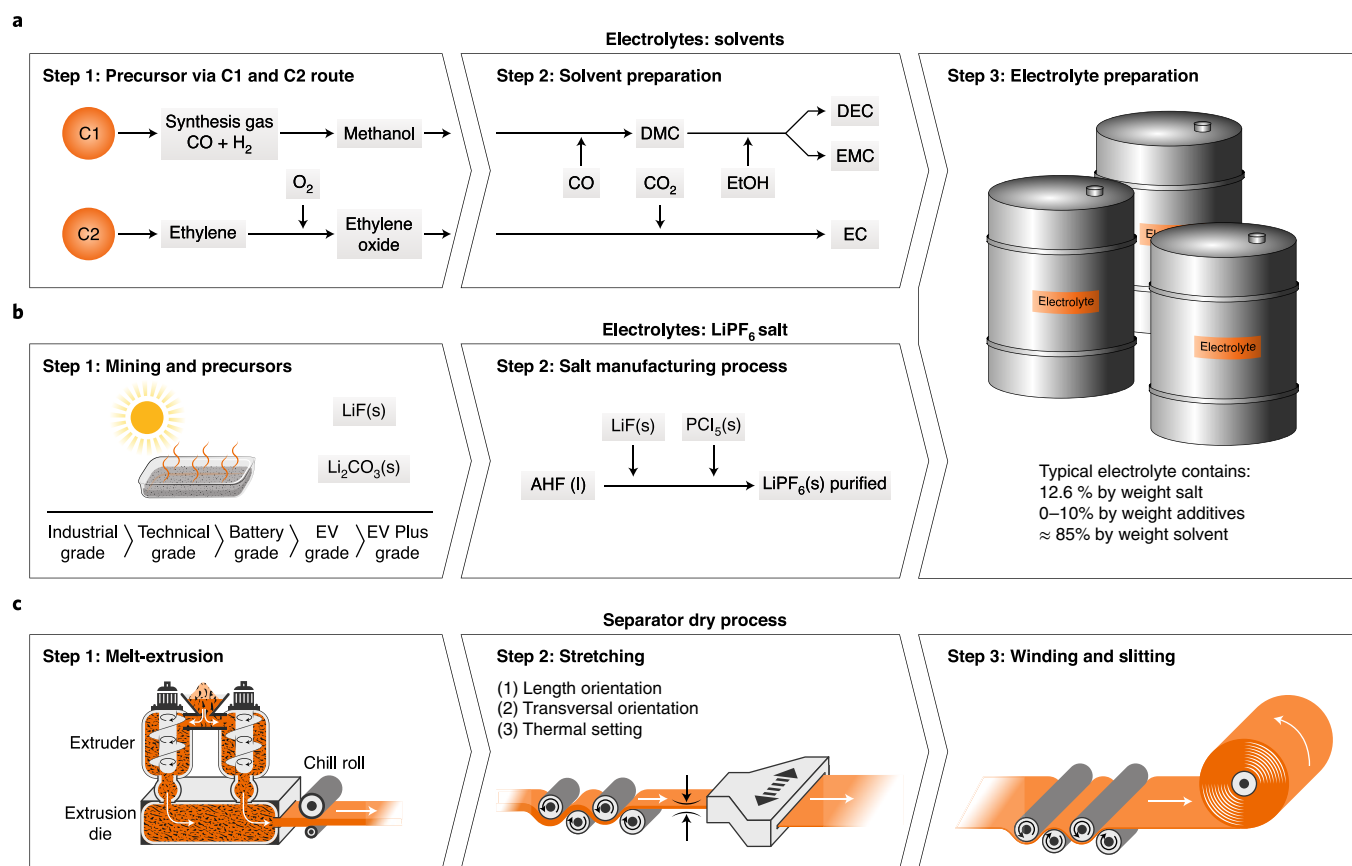


Fig. 5 | Production processes for separators and electrolytes. a,b, Schematic processes of electrolyte component manufacturing and mixing starting for solvents from C1 and C2 synthesis routes in bulk chemistry production (**a**) and for salts from brine (**b**). Not shown are production processes for hundreds of additives, which would go beyond the scope of this paper. **c,** Schematic showing the separator dry process. CO, carbon monoxide; CO₂, carbon dioxide; EtOH, ethanol; DMC, dimethyl carbonate; DEC, diethyl carbonate; EMC, ethyl methyl carbonate; EC, ethylene carbonate; C1, one-carbon synthesis route; C2, two-carbon synthesis route; AHF, anhydrous hydrogen fluoride.

by a factor of two or more, currently limiting non-woven separators in LIBs to niche markets⁷⁶. Microporous membranes and non-woven mats may be additionally coated with ceramic⁷⁷, for example, Al₂O₃ at additional costs of 0.3 US\$ m⁻². Ceramic coating supports higher temperature tolerance and, thus, safety, which is increasingly demanded for large battery cells for the car industry⁷⁶. Further, ceramic particles also improve the wettability of the separator⁷⁶. Despite their multilayer architecture, the thickness of separators is 20 μm or less, enabling high volumetric energy and power density⁷⁶. It is common understanding that those separators containing (basic) ceramic particles can absorb HF generated during cycling and, thus, may support longer cycle life.

To replace unsafe organic solvents in the conventional liquid-electrolyte–separator set-up, so-called solid electrolytes, serving as both separator and electrolyte, are attracting increasing interest^{40,78}. Solid electrolytes can be classified into two major categories: solid polymer electrolytes (SPEs) and inorganic solid electrolytes either having a crystalline, glass or glass-ceramic structure⁴⁰. It is believed that solid electrolytes will enable rechargeable Li-metal anodes, which can boost the energy density at cell level⁷⁹. SPEs based on poly(ethylene oxide) (PEO) together with Li-metal anode and LFP cathode are already used as traction cells in car-sharing programs. Due to the poor room-temperature conductivity of SPEs, the battery packs are operated at 60–80 °C, which reduces the usable electric energy at pack level⁸⁰. Further, PEO has a low oxidative stability, limiting the usable cathode chemistry to materials operating below 4 V (ref. ⁸¹).

Contrary to common belief in the research community, some inorganic solid electrolytes have even higher room-temperature conductivities than conventional liquid electrolytes, but often have smaller electrochemical stability windows⁴⁰. The major challenge is to conform the solid active material/solid electrolyte interface and, thus, maintain the proceeding interfacial reactions during cycling, in particular as shape-change and insertion electrode materials continuously swell and shrink during operation⁴⁰. Until recently, inorganic solid electrolytes were only commercially used in thin-film batteries based on lithium phosphorous oxynitride (LIPON)⁸¹. Still, their high cost hampers a wider market penetration of these thin-film cells⁴⁰. Considering the already very low cost liquid electrolyte and separator as a benchmark, there is not much room for reducing the material processing cost of solid electrolytes⁴⁰. A cost disadvantage certainly lies in the typically high lithium content in the solid electrolytes discussed, whose development is still at an early stage. The future role of Li-metal-based ASSBs and the potential displacement of LIBs greatly depends on major technological breakthroughs.

Cell chemistries for automotive batteries

Table 1 gives an overview of the cell chemistries (anode and cathode combination) and characteristics for various cells and batteries as well as estimated driving ranges of commercial BEVs from past and current production^{4,11,82}. In recent years, the energy content (or ‘battery size’) of EVs has continuously increased to currently up to 100 kWh per vehicle, and higher driving ranges (over 300 km)

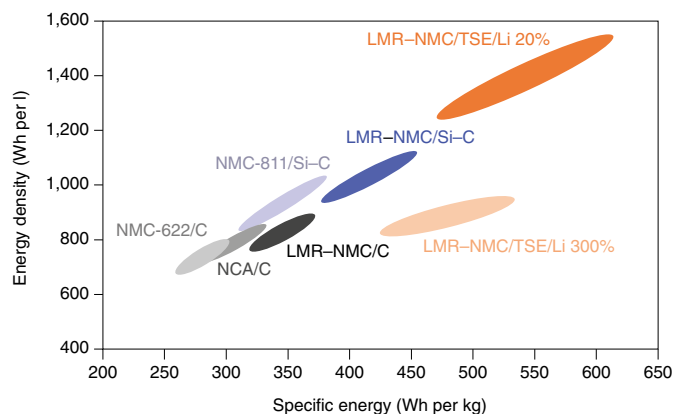


Fig. 6 | Energy density versus specific energy diagram of different cell chemistries at electrode stack level. Schematic illustration of energy contents per volume and per weight in the form of ellipses for various material combinations. The state-of-the-art LIB chemistry is represented by NMC-622/graphite and NCA/graphite. The colour code addresses the anode chemistry with grey, blue and orange representing carbon, silicon and Li-metal based chemistries, respectively. The ellipses are based on the calculation of different values for varying positive electrode thicknesses (50–150 μm). The energy content was calculated using a spreadsheet-based calculation tool, using physical properties (for example, material densities, electrode porosities, film thicknesses and electrode composition) and electrochemical properties (for example, practical discharge potential and redox potential) as input parameters. C, graphite; Si-C, silicon-carbon composite (20 wt% Si); NMC, $\text{LiNi}_{1-x-y}\text{Co}_x\text{Mn}_y\text{O}_2$; NCA, $\text{LiNi}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$; LMR-NMC, lithium- and manganese-rich NMC; TSE, thiophosphate-based solid electrolyte (for example, $\text{Li}_7\text{P}_3\text{S}_{11}$); Li, lithium metal (with 20% or 300% excess).

have therefore become attainable. As a result of enhanced ranges, car manufacturers are willing to compromise on the targeted cycle number (1,300–2,000) (ref. ⁸), as the electricity throughput of a battery (around 40 MWh) to drive an EV for a certain accumulated tyre mileage (for example, 200,000 km with an energy consumption of 20 kWh per 100 km) is more critical than the cycle number. However, the cycle number, and not the total electricity throughput, is often seen as critical parameter, which is still a popular misconception.

Since its market introduction, the energy content of LIBs has increased remarkably as a result of numerous material and engineering measures⁶. Prismatic, cylindrical or pouch cell designs are used in automotive applications, even though the highest specific energy and energy density at cell level is currently achieved by small cylindrical 18650-type cells (around 250 Wh kg^{-1} and 670 Wh l^{-1}) (refs ^{6,40}). The LIB technology does not rely on specialized cell chemistries, so various material combinations can be applied to meet desired requirements. This becomes clear when looking at the material combinations used in commercial cells (Table 1). The majority of EV cells are based on synthetic graphite anodes, while recently, small amounts of silicon were added to the anode for some cells. If silicon can be successfully established in the anode, without shortening cycle life excessively, further energy density improvements could be realized.

Various cathode active materials or combinations are commercially applied, in particular Ni-rich layered cathodes, which are commonly blended with LMO for high power density and safety. With respect to further energy density improvements, a further decrease of the LMO content is, therefore, an obvious strategy. Moreover, the Ni content in layered cathodes has continuously increased to enhance capacity. Currently, NCA, NMC-532 and NMC-622 can be considered as state-of-the-art cathode materials. Due to its

relatively low volumetric capacity, lithium iron phosphate (LFP), which is used in passenger cars in China at a large scale, is a candidate for replacement by higher capacity layered oxides. Due to the robustness of LFP, there remains an opportunity in heavy-duty applications, such as buses and trucks, where the material's long cycle life and good rate capability come into play.

Figure 6 displays the electrode stack level energy content of various cell chemistries, including state-of-the-art chemistries and future LIB and Li-metal technologies. The ellipses are based on the calculation of different values for positive electrode thicknesses, ranging from 50 μm (corresponding to lower energy content) up to 150 μm (corresponding to higher energy content). The utilization of thicker and/or denser electrodes is a suitable strategy to increase the energy, if the power demands can still be fulfilled at the same time. Currently, coating thicknesses of around 65–80 μm are used for high energy cells. Further energy improvements can be achieved using high-capacity anode and cathode materials, that is, Si and LMR-NMC or NMC-811, or by moving to the Li-metal technology. As proposed by various researchers and companies, lithium metal anodes will most likely only be realized in conjunction with solid electrolytes, which are used to provide sufficient safety that cannot be accomplished with current liquid electrolytes, although ionic liquid-based electrolytes show some promise⁸³. The overall gain in energy for Li-metal batteries will also be strongly influenced by the excess of Li-metal (Fig. 6), which might vary between 20–300%, depending on the extent of parasitic, Li-consuming reactions reducing the active lithium content during operation.

ASSBs involving an inorganic solid electrolyte might constitute the first disruptive innovation in the field of rechargeable Li battery technology since its introduction⁷⁸. In comparison to LIBs, a leap towards higher energy densities could be realized if a Li-metal anode can be safely and reversibly utilized. It is also envisioned that multiple bipolar ASSB stacks can be connected in series without the need for individual cell housings, which would in turn increase the active:inactive ratio in terms of both volume and weight and, thus, increase the overall energy density of ASSBs⁸⁴. However, bipolar configuration also means that there is no single cell management control and thus very careful cell capacity balancing is necessary. Other promises of ASSBs include an improved high-power capability, enhanced thermal stability, and extended cycle and calendar life⁴⁰. Currently, thio-LISICON-type solid electrolytes from the $\text{Li}_2\text{S-P}_2\text{S}_5$ system and oxidic electrolytes such as NASICON-type $\text{Li}_{1-x}\text{Al}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ or garnets such as $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ are among the most promising materials classes for advanced ASSBs^{78,81}. Toyota have announced that they will introduce ASSB-powered EVs with sulphide-based solid electrolytes by 2022⁸⁵. However, each solid electrolyte chemistry faces individual scientific and technological challenges, including limited chemical stability towards other cell components, instability towards exposure to atmosphere or moisture, difficulties in processability, and high interfacial resistances. Furthermore, the cycling stability against Li-metal still has to be proven and the solid/solid interfaces have to be tailored carefully to achieve suitable performance. Besides ASSBs, lithium-sulphur (Li-S) batteries were considered as an alternative high-energy battery for automotive applications, mainly due to their exceptionally high theoretical specific energy at the material level and the low cost of sulphur^{86–88}. These theoretical values often deviate markedly from practical values, as the large weight and volume fraction of inactive battery components such as the electrolyte or conductive additive as well as the volume expansion of the active materials are not taken into account^{88,89}. For automotive applications, energy density (Wh l^{-1}) is considered more critical than specific energy (Wh kg^{-1}). Optimized Li-S cells could outperform advanced LIBs in terms of Wh kg^{-1} , but not Wh l^{-1} , and would then not be attractive for EVs, but rather for applications with need for high specific energies (such as aviation, military applications)⁹. Also, lithium-air

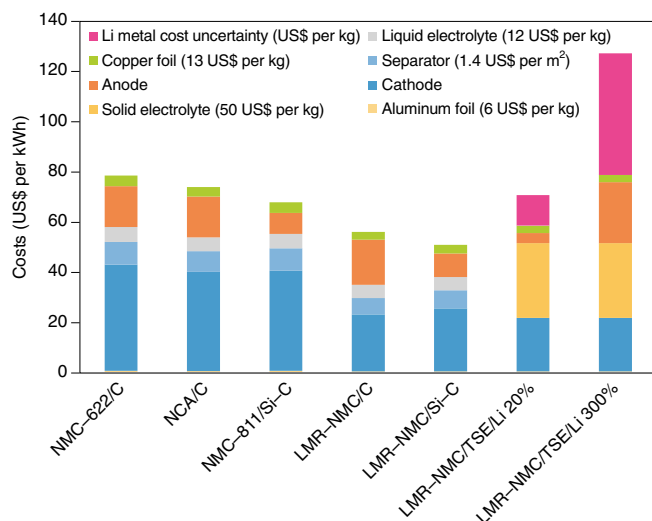


Fig. 7 | Cost estimations for different cell chemistries at electrode stack level. Calculations are based on a positive electrode thickness of 65 μm , a negative:positive capacity ratio of 1.10 and a production scrap rate of 5%. The following material porosities were used: graphite and positive electrodes (30%), Si-C composite electrodes (50%; 20 wt% silicon), separator (39%), solid electrolyte and Li-metal anode (0%). The following material costs were used: C, graphite (19 US\$ per kg); Si-C, silicon-carbon composite (20 wt% silicon; 25 US\$ per kg); NMC ($\text{LiNi}_{1-x-y}\text{Co}_x\text{Mn}_y\text{O}_2$; NMC-622: 25 US\$ per kg; NMC-811, 24 US\$ per kg); NCA, $\text{LiNi}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$ (26 US\$ per kg); LMR-NMC, lithium or manganese-rich 'layered-layered' transition metal oxides (20 US\$ per kg); TSE, thiophosphate-based solid electrolyte (for example, $\text{Li}_7\text{P}_3\text{S}_{11}$) (50 US\$ per kg). Li, lithium metal: thin Li-foil, 20 μm , 20% excess Li (250–1,000 US\$ per kg; 250 US\$ per kg indicated as anode cost plus up to 300% cost uncertainty); thick Li-foil, 200 μm , 300% excess Li (100–300 US\$ per kg; 100 US\$ per kg indicated as anode cost plus up to 200% cost uncertainty). Process and overhead cost, as reported in previous studies^{96–98} have not been considered in this calculation. The calculation takes into account the anode, cathode, electrolyte and separator.

batteries have been considered as a potential successor of the LIB, owing to the high theoretical energy density on the material level. On a closer look at the electrode⁸⁹ or system level⁹⁰, which includes the mass and volume of the required inactive and auxiliary components, the projected energy density is significantly reduced. Lithium-air batteries are currently still at a very early stage of development. The many technical challenges are not expected to be overcome within the next decade and it remains to be seen whether this technology will find application in electric vehicles in the far future.

Fast-charging capabilities for automotive batteries have drawn much focus, as this factor is crucial for consumer acceptance. The target of various car manufacturers is to achieve 80% state-of-charge (SOC) within 5–20 minutes⁹². To reach this goal, the cells and battery pack have to be specifically designed to meet the demanding charging requirements, at the expense of the energy density, for example by using higher amounts of LMO in the cathode or by additional measures for cell cooling. In addition, the risk of increased capacity fading and more severe safety considerations have to be taken into account. For example, to fully charge a 60 kWh battery (with around 380 km range; Table 1) within 20 minutes (more than 3C rate), a charging power of 180 kW would be required, which, in turn, leads to a driving range gain of 15 km per minute of charge. In comparison, an internal combustion engine-driven vehicle (ICEVs) can be refuelled with an equivalent rate of 12C. In contrast, a rate of 0.3–0.5C is typically sufficient for battery discharge.

The cost of EV battery packs has continuously declined in recent years. The price of LIB battery packs has fallen below 300 US\$ kWh⁻¹ and is expected to decrease further⁹¹. To achieve cost competitiveness with combustion-engine-driven vehicles, the pack cost needs to fall below 120–150 US\$ kWh⁻¹ (ref. ⁹¹). Cost savings have been partly realized through up-scaling and automation of the battery production in large industrial plants. The large investments for a large battery factory make market entry for new players very difficult. Furthermore, cost competitiveness with established manufacturers is difficult to achieve. Traditional car manufacturers are currently not likely to commit to their own battery-cell production, while start-up companies, joint ventures or automotive component suppliers are more likely to do so⁹². Apart from economies of scale, further cost savings may be realized by changing the cell type, for example, by enlarging the dimensions of cylindrical cells from 18650 to 21700 format, so that the active:inactive material volume ratio is increased.

The main cost reduction drivers of EV cells are related to manufacturing and production improvements, such as increasing yield and throughput rate, as well as decreasing material cost per kWh. Furthermore, producing higher volumes of materials will result in lower costs per unit. Nevertheless, there are various risks when targeting such low cost levels, particularly with respect to the raw material cost of cobalt, nickel, lithium and copper⁸². In October 2017, copper prices rose to their highest level in three years. The copper price is however not critical at cell level, since only small quantities are needed as current collector⁹³. However, with regard to the cell hardware, wiring system components in the EV as well as the infrastructure for fast charging, copper can be regarded as a strategic material, meaning that it is a strategic requirement to recycle copper and to promote eco-designed products with limited copper usage⁹³. Supply risks of LIB raw materials have been evaluated in various studies, pointing out that the highest risk category with respect to material reserves and supply reliability includes cobalt and lithium^{59,94,95}. However, concerns about Li shortage are not shared by all experts in the field.

Figure 7 illustrates projected cost per kWh for various cell chemistries, which already have been considered for the energy calculations at stack level (see Fig. 1b). With respect to LIB technologies, it is obvious that for a given volume the cost per kWh decreases with increasing energy content. The estimated cost for Li-metal-based ASSBs, despite having higher energy densities, is significantly higher compared to the high-energy LIB technologies, which is mainly due to the high cost of Li-metal (250–1,000 US\$ kg⁻¹ for 20 μm thickness or 100–300 US\$ kg⁻¹ for 200 μm thickness Li foil). When using a thicker Li-metal foil, the Cu current collector foil can theoretically be omitted, which would result in a marginal cost saving of roughly 2 US\$ kWh⁻¹. The high Li-metal costs also make Li-S and Li-O₂ based chemistries more expensive than regularly claimed. Nevertheless, it has to be kept in mind that the cost for lithium metal foil strongly depends on the targeted thickness and processing technique, such as extrusion, rolling or even vapour deposition or evaporation, as indicated by the Li metal price uncertainty in Fig. 7. Due to unavailable data, the cost for the solid electrolyte has been roughly estimated as 50 US\$ kg⁻¹. Furthermore, the commonly claimed safety benefits of ASSBs also have to be verified in large-scale cells at various states of health.

Conclusions and future perspectives

It is widely expected that rechargeable batteries will require energy contents of around 235 Wh kg⁻¹ and 500 Wh l⁻¹ at pack level to achieve a driving range beyond 500 km. Currently, up to 140 Wh kg⁻¹ and 200 Wh l⁻¹ can be achieved at battery pack level. To achieve these demanding energy targets, several strategies have been followed from material to pack level, such as the implementation of a higher Si-content in high-energy anodes and the increase of the

Ni content in layered cathode materials. Advances in the attainable driving range have been realized in recent years, as the energy content has grown to 100 kWh per EV. With increasing driving range per cycle, car manufacturers are in turn willing to compromise on the attainable cycle number, as the overall energy throughput of a battery is more crucial. Energy improvements will only be possible if the power capability is not severely compromised, as the feasibility of fast charging (that is, 80% SOC in 20 min) will be critical for user acceptance.

Battery pack cost needs to fall below 125 US\$ kWh⁻¹ to achieve cost competitiveness with ICEVs. At the same time, ICEVs are likely to become increasingly expensive due to stricter CO₂ and NO_x emission regulations. Currently, the battery costs range from 100–170 US\$ kWh⁻¹ and 220–250 US\$ kWh⁻¹ at the cell and pack level, respectively, which strongly depends on the chosen cell chemistry. It can be assumed that the cost target of 125 US\$ kWh⁻¹ at pack level will be achieved by 2022–2025, while the cost reduction drivers are economies of scale and a higher degree of automation in production. Cobalt is considered as the most critical raw material with respect to cost, material reserves and supply reliability, while concerns about Li availability are not shared by all experts in the field.

We expect a further improvement of today's LIBs with respect to both energy and cost. Currently, due to a favourable mix of numerous parameters, the LIB is superior to alternative cell technologies, where research is driven by individual (and often theoretical) advantages over current LIBs. It is also possible that new LIB cell chemistries — with even higher capacities than projected on today's material roadmaps — will be developed in the future.

To achieve a comparable mix of advantageous properties with alternative chemistries will certainly take time, if it can ever be realized. Therefore, LIBs are likely to still dominate automotive applications within the coming decade. Nevertheless, it also is clear from today's perspective that there will not be only one universal LIB technology for all applications, but rather a variety of chemistries and configurations for specialized applications. Considering the long development cycles of the automotive industry, it is currently unlikely that LIB technology will be replaced by any 'next generation' technology such as Li-metal based chemistries (ASSBs, Li-S, Li-O₂) or by technologies beyond lithium (such as Na- or Mg-based chemistries) in the next decade. Considering the large financial investments already committed to the large-scale production of LIBs worldwide, it becomes increasingly more difficult for emerging battery technologies to be cost competitive. From today's perspective, it is only reasonable to state that the next generation of automotive batteries will be a further improved version of current LIBs.

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Competing interests

The authors declare no competing interests.

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